



Mansoura University
Faculty of Computers and Information
Department of Information System



[IS422P] Data Mining

Grade:4 Th

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CLASSIFICATION (PART I)



AGENDA



The Basics

What is Classification?
General Approach



Decision Tree Induction

The Algorithm
Attribute Selection Measures
Tree Pruning
Extracting Rules from Decision Trees



Bayes Classification

Bayes' Theorem
Naïve Bayesian Classification



Lazy Learners

K-Nearest Neighbor Classifiers



Regression analysis

Linear regression



Model Evaluation

Metrics for Evaluating Classifiers Performance
Cross-Validation
Bootstrap



Improving Classification Accuracy

Bagging
Boost and AdaBoost



THE BASICS

WHAT IS CLASSIFICATION

- Motivation: Prediction
 - Is a bank loan applicant “safe” or “risky”?
 - Which treatment is better for patient, “treatmentX” or “treatmentY”?
- Classification is a data analysis task where a model is constructed to predict class labels (categories)



THE BASICS

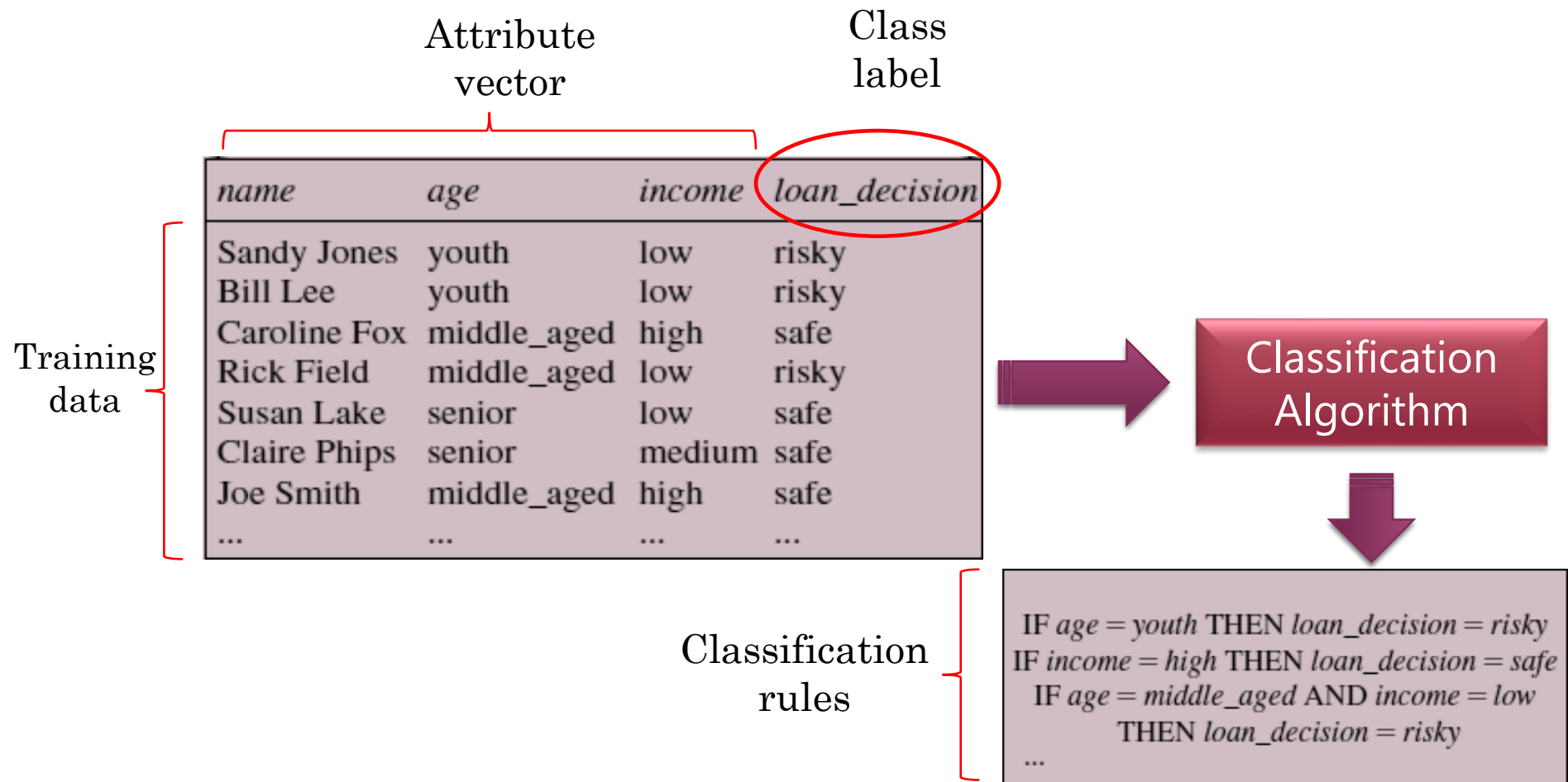
GENERAL APPROACH

- A two-step process:
- Learning (training) step → construct classification model
 - Build classifier for a predetermined set of classes
 - Learn from a training dataset (data tuples + their associated classes) → Supervised Learning
- Classification step → model is used to predict class labels for given data (test set)



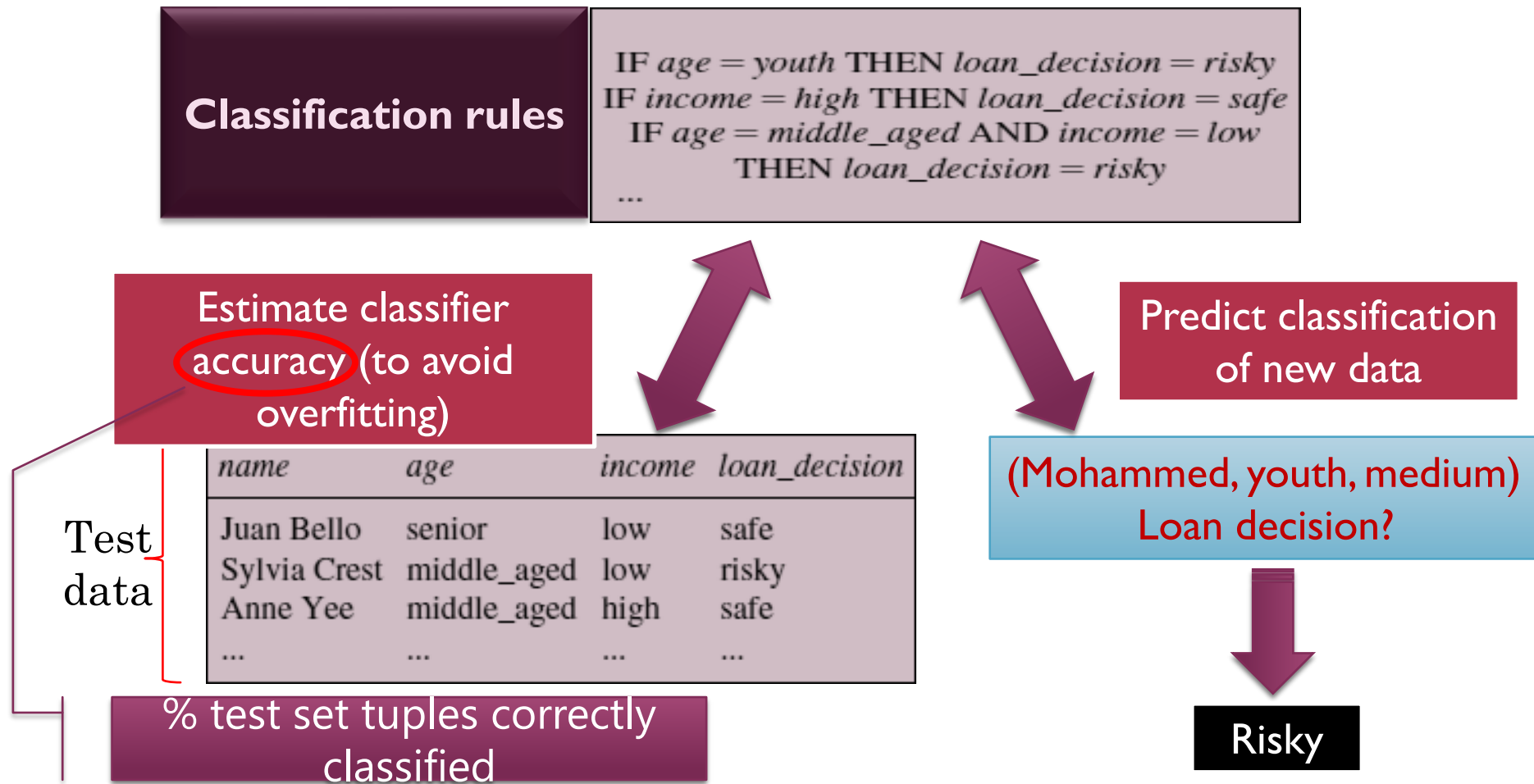
THE BASICS

GENERAL APPROACH



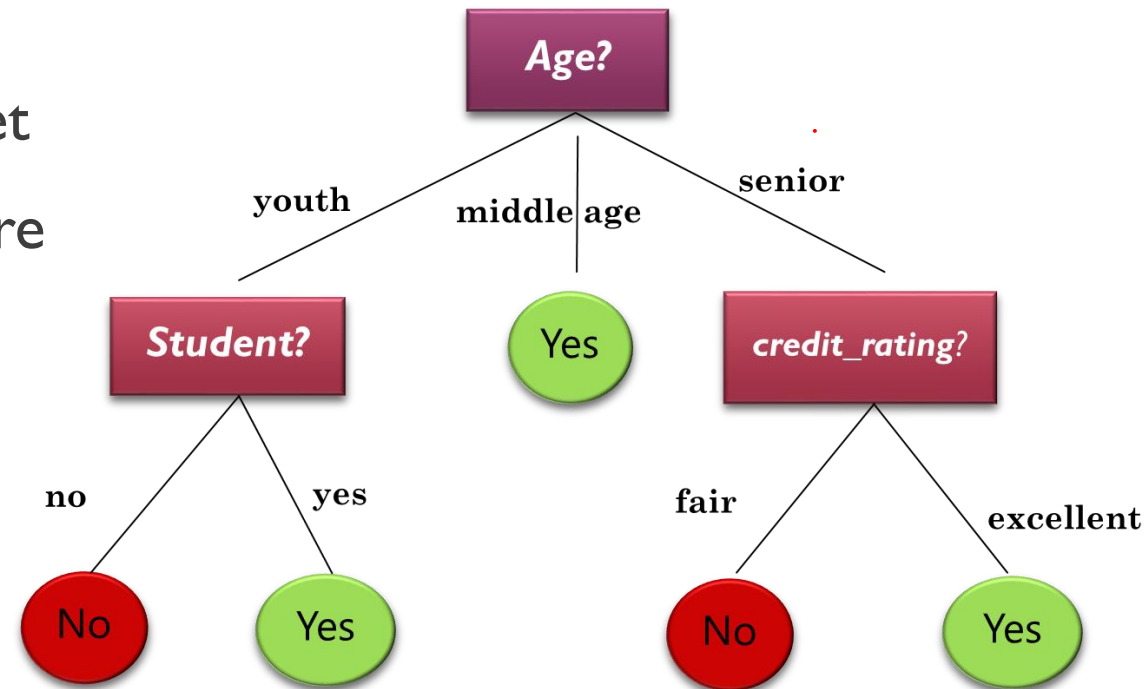
THE BASICS

GENERAL APPROACH



DECISION TREE INDUCTION

- Learning of decision trees from training dataset
- Decision tree → A flowchart-like tree structure
 - Internal node → a test on an attribute
 - Branch → a test outcome
 - Leaf node → a class label
- Constructed tree can be binary or otherwise



DECISION TREE INDUCTION

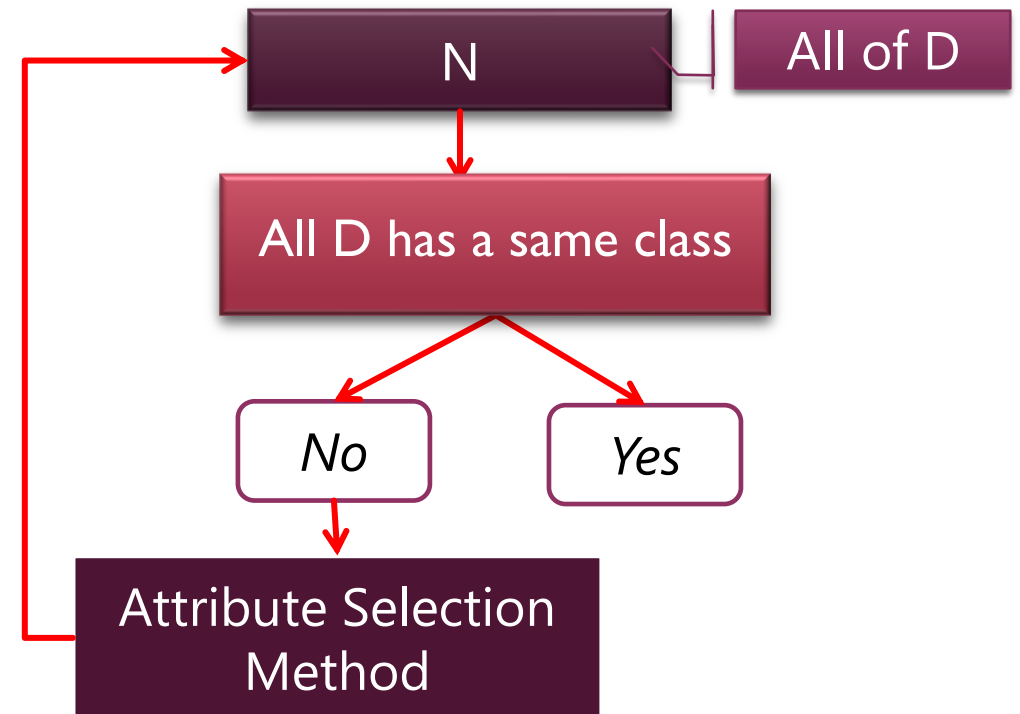
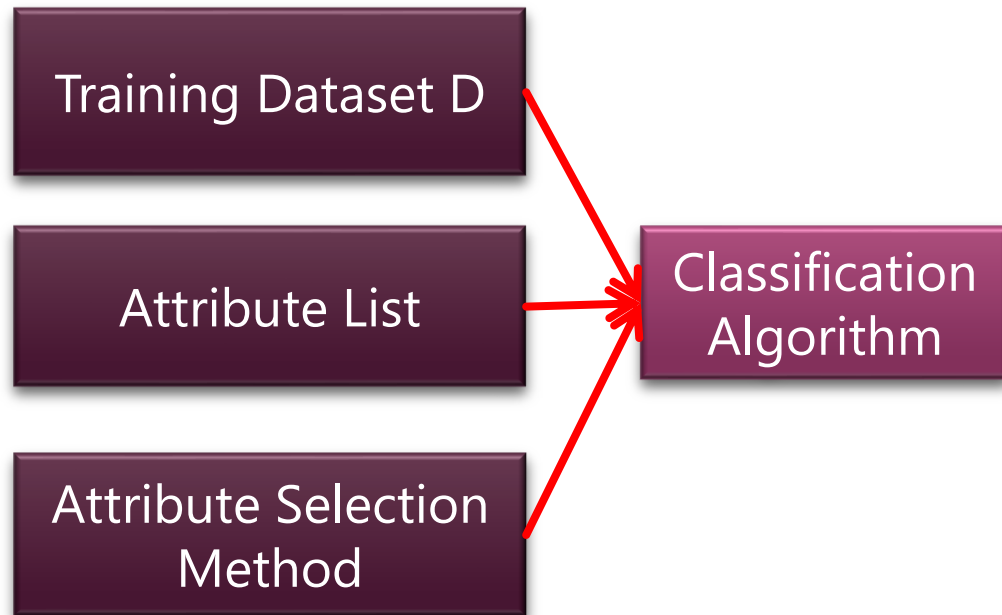
Benefits

- No domain knowledge required
- No parameter setting
- Can handle multidimensional data
- Easy-to-understand representation
- Simple and fast



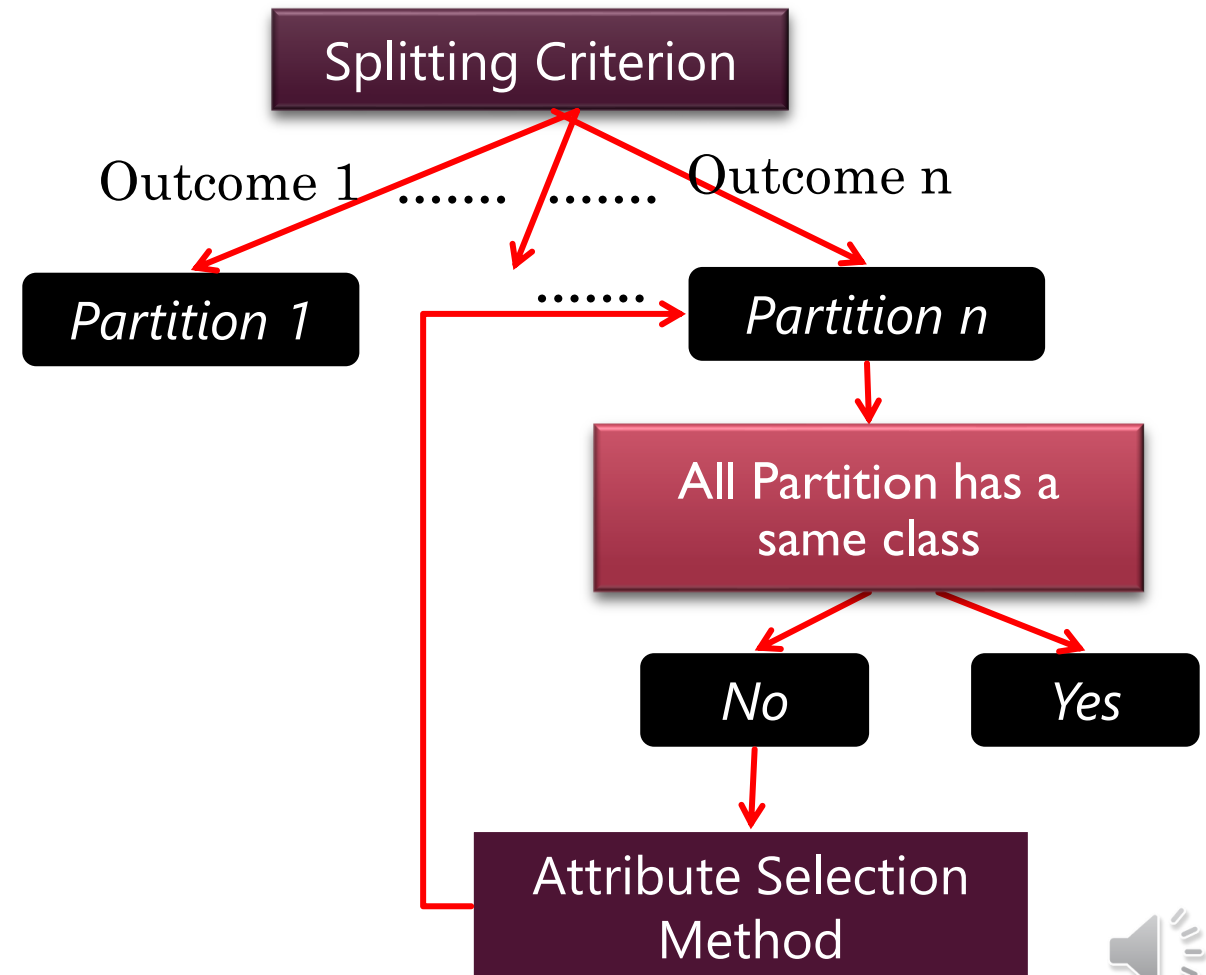
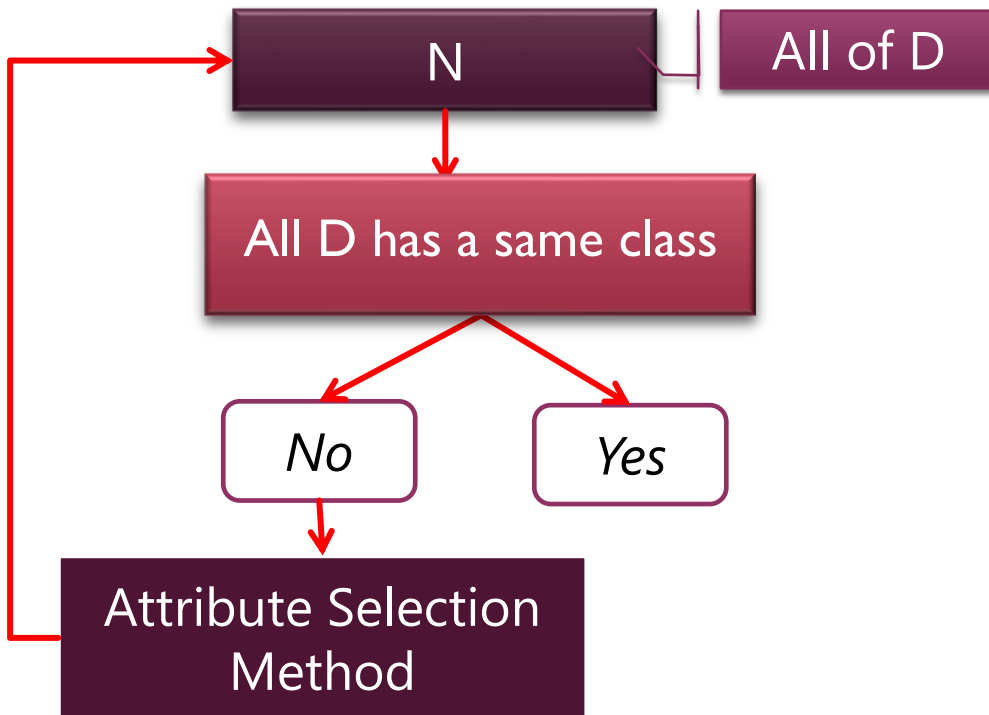
DECISION TREE INDUCTION

THE ALGORITHM



DECISION TREE INDUCTION

THE ALGORITHM



DECISION TREE INDUCTION

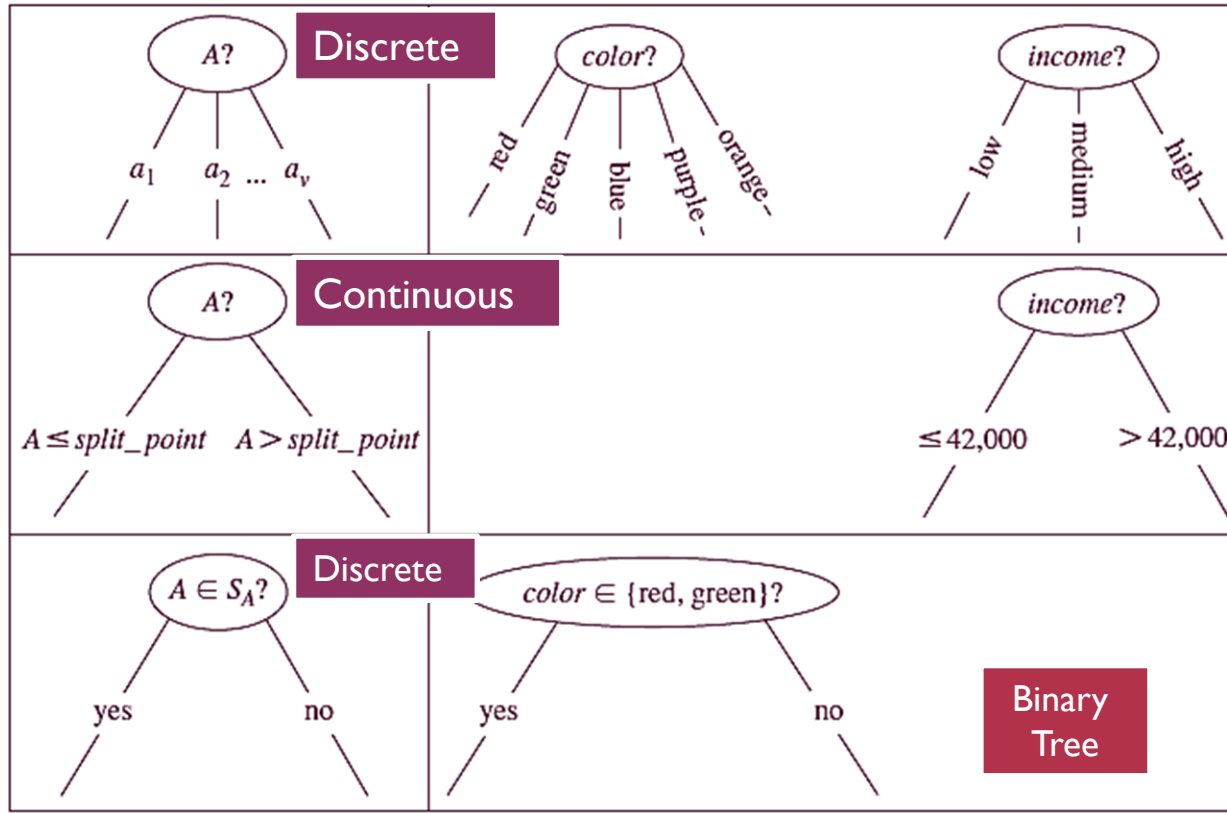
THE ALGORITHM

Splitting Attribute

Splitting Criterion

Partitioning scenarios

Examples



Outcome 1 Outcome n

Partition 1

Partition n

All Partition has a same class

No

Yes

Attribute Selection Method



DECISION TREE INDUCTION

THE ALGORITHM

- **Splitting Criterion is a test:**
 - Which attribute to test at node $N \rightarrow$ What is the “best” way to partition D into mutually exclusive classes
 - which (and how many) branches to grow from node N to represent the test outcomes
- Resulting partitions at each branch should be as “pure” as possible
 - A partition is “pure” if all its tuples belong to the same class
- When attribute is chosen to split training data set, it's removed from attribute list



DECISION TREE INDUCTION

THE ALGORITHM

- **Terminating conditions**

- All the tuples in D (represented at node N) belong to the same class
- There are no remaining attributes on which the tuples may be further partitioned
 - majority voting is employed $\rightarrow$ convert node into a leaf and label it with the most common class in data partition
- There are no tuples for a given branch
 - a leaf is created with the majority class in data partition



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES

- **Attribute selection measure** → a heuristic for selecting the splitting criterion that “best” splits a given data partition into smaller mutually exclusive classes
- Attributes are ranked according to a measure
 - attribute having the best score is chosen as the splitting attribute
 - split-point for continuous attributes
 - splitting subset for discrete attributes with binary trees
- Measures: Information Gain, Gain Ratio, Gini Index



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES

Information Gain

- Based on *Shannon's information theory*
- Goal is to **minimize the expected number of tests needed to classify a tuple**
 - guarantee that a simple tree is found
- Attribute with the highest information gain is chosen as the splitting attribute
 - minimizes information needed to classify tuples in resulting partitions
 - reflects least “impurity” in resulting partitions



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES

- Given m class labels ($C_i, i = 1$ to m)
- Expected Information needed to classify a tuple in D
- $\text{Info}(D) = \text{entropy} = - \sum_{i=1}^m p_i \log_2(p_i)$
- $p_i \rightarrow$ probability that an arbitrary tuple in D belong to class C_i

$$p_i = \frac{|C_{i,D}|}{|D|}$$

- $C_{i,D} \rightarrow$ set of tuples having class label C_i in partition D



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES

- How much more Information would be needed after Partitioning to arrive at a “pure” classification”
 - Expected information required to classify a tuple from D based on the partitioning by attribute A:
 - $info_A(D) = \sum_{j=1}^v \frac{|D_j|}{D} \times info(D_j)$
 - The smaller the expected information still required, the greater the purity of the partitions



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES

- **Information gain** is the different between the original information required (based on proportion of classes) and the new requirement (after partitioning on A)
- $Gain(A) = info(D) - info_A(D)$
- $Gain(A)$ tells you how much would be gained by branching on A
 - Expected reduction in the information requirement caused by knowing the values of A
 - Attributes A with the highest $Gain(A)$ is chosen as the splitting attribute at node N



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES

department	age	salary	status	count
sales	Middle aged	medium	senior	30
sales	youth	low	junior	30
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systems	youth	medium	junior	20
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$$CI(\text{Senior}) = 80, C2(\text{Junior}) = 120$$

$$\begin{aligned} \text{info}(D) = \text{entropy} &= -\sum_{i=1}^m p_i \log_2 p_i \\ &= -\frac{80}{200} \log_2 \frac{80}{200} - \frac{120}{200} \log_2 \frac{120}{200} = 0.97 \end{aligned}$$

$$\text{info}_A(D) = -\sum_{j=1}^n \frac{D_j}{D} \text{info}(D_j)$$

Department: Sales = 100, system = 50, marketing = 30, secretary = 20

Info_{department} =

$$\begin{aligned} &\frac{100}{200} \left(-\frac{30}{100} \log \frac{30}{100} - \frac{70}{100} \log \frac{70}{100} \right) + \frac{50}{200} \left(-\frac{30}{50} \log \frac{30}{50} - \frac{20}{50} \log \frac{20}{50} \right) \\ &+ \frac{30}{200} \left(-\frac{10}{30} \log \frac{10}{30} - \frac{20}{30} \log \frac{20}{30} \right) + \frac{20}{200} \left(-\frac{10}{20} \log \frac{10}{20} - \frac{10}{20} \log \frac{10}{20} \right) \\ &= 0.92 \end{aligned}$$



DECISION TREE INDUCTION

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$$\text{Gain}_{\text{department}} = 0.97 - 0.92 = 0.05$$



DECISION TREE INDUCTION

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$$\text{Gain}_{\text{department}} = 0.97 - 0.92 = 0.05 \text{ bits}$$

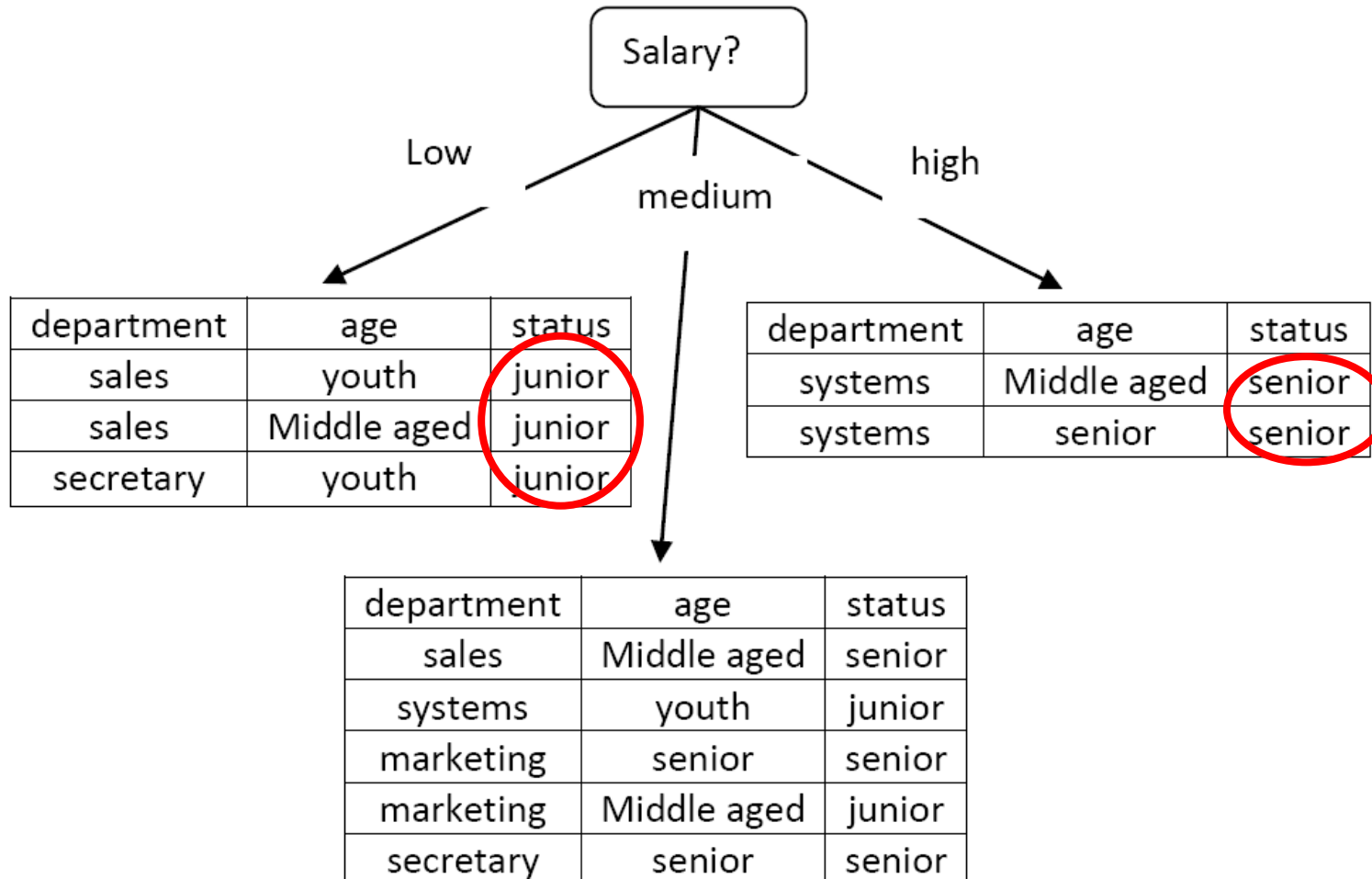
$$\text{Gain}_{\text{Age}} = 0.97 - 0.55 = 0.42 \text{ bits}$$

$$\text{Gain}_{\text{Salary}} = 0.97 - 0.45 = 0.52 \text{ bits}$$



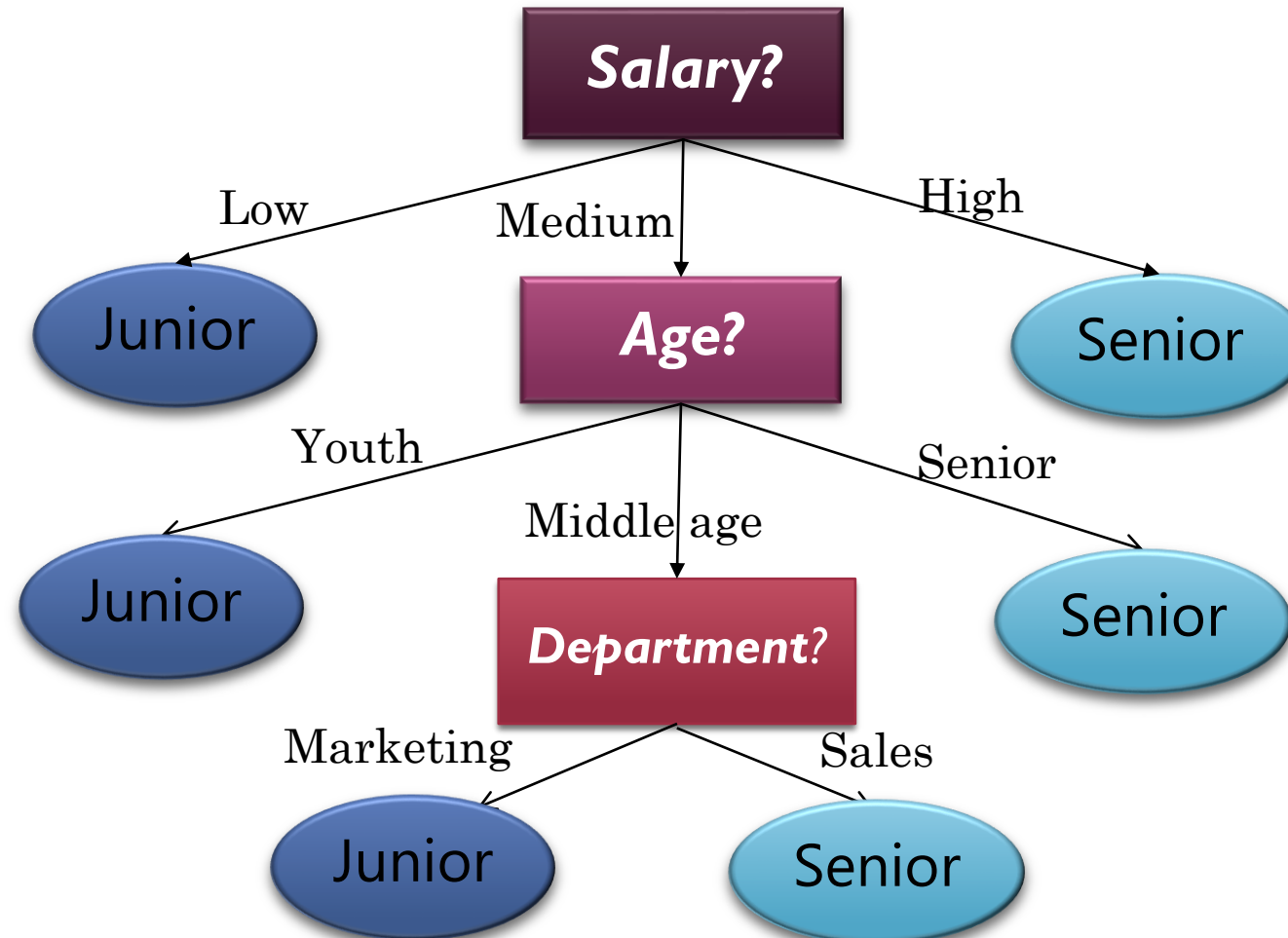
DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES



DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES

*Information gain for **continuous attributes***

1. Sort values in increasing order
2. Each *midpoint* between two adjacent values can serve as *split-point*
3. Split-point between two values v_i and $v_{i+1} = \frac{v_i + v_{i+1}}{2}$
4. For each split-point, evaluate $info_A(D)$ with the number of partitions = 2 ($A \leq split-point$ & $A > split-point$)



DECISION TREE INDUCTION

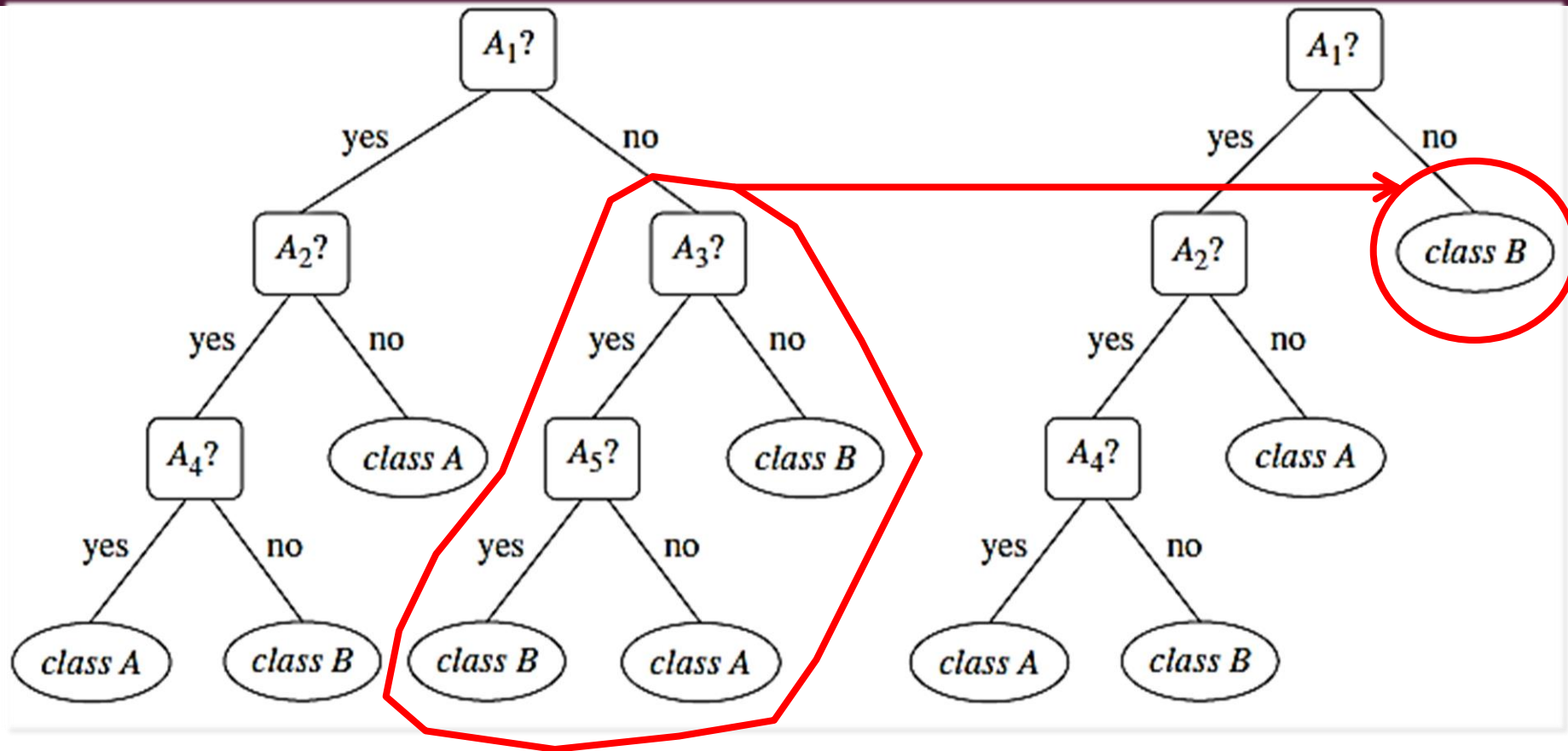
TREE PRUNING

- Data may be *overfitted* to dataset anomalies and outliers
- **Pruning** removes the least reliable branches
 - DT becomes less complex
- Prepruning → statistically assess the *goodness of a split* before it takes place
 - hard to choose *thresholds* for statistical significance
- Postpruning → remove sub-trees from already constructed trees
 1. remove sub-tree branches and replace with leaf node
 2. leaf is labeled with most frequent class in sub-tree



DECISION TREE INDUCTION

TREE PRUNING



DECISION TREE INDUCTION

RULE EXTRACTION FROM A DECISION TREE – WHAT ARE RULES?

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RI: IF salary =medium AND age = youth
THEN Status = Junior

$$coverage(R) = \frac{n_{covers}}{|D|}$$

$$accuracy(R) = \frac{n_{correct}}{n_{covers}}$$

- coverage(RI) = 20/200=10% and
- accuracy(RI)= 20/20 = 100%.

X: (Department = system age = youth, salary = low)

?



DECISION TREE INDUCTION

RULE EXTRACTION FROM A DT – RESOLVING RULES CONFLICTS

Rules conflicts are the result of a tuple firing more than one rule with different class predictions

Two resolution strategies

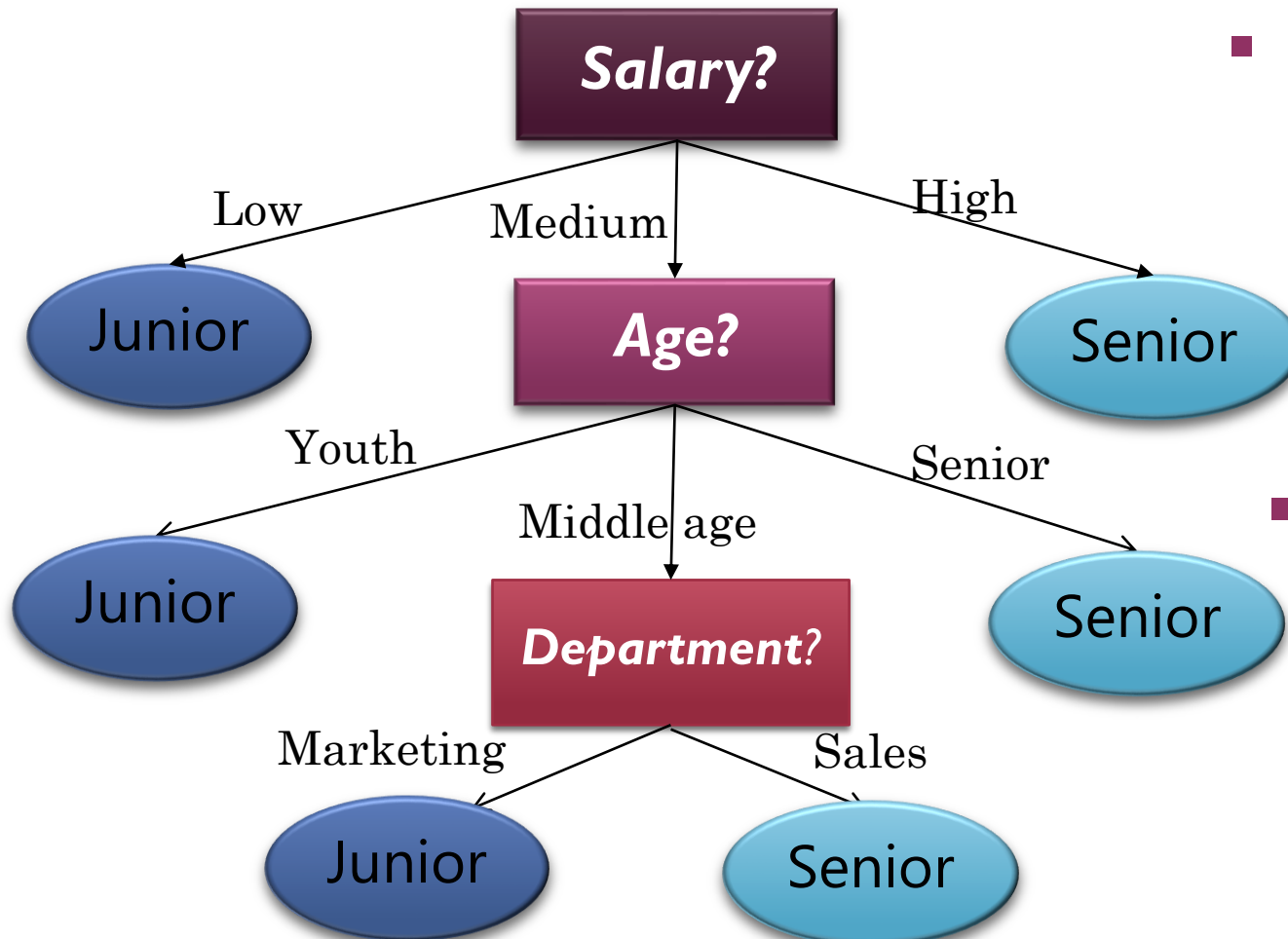
- **Size Ordering** → rule with **largest antecedent** (toughest) has highest priority fires and returns class prediction
- **Rule Ordering** → rules **prioritized apriori** according to
 - **Class-based ordering** → decreasing importance (most frequent are highest – order of prevalence)
 - **Rule-based ordering** → measures of rule quality (e.g. accuracy, size, domain expertise)

Fallback (default) rule when no rules are triggered



DECISION TREE INDUCTION

RULE EXTRACTION FROM A DECISION TREE



- Create one rule for each path from root to leaf in the decision tree
 1. Each splitting criterion is ANDed to form rule antecedent (IF)
 2. Leaf node holds class prediction (THEN)
- RI: IF salary =medium AND age = youth THEN Status = Junior

Can the rules resulting from decision trees have conflicts?



QUESTION?

NEXT

